
Feature Dimension Reduction: PCA & LDA

HYEONWOO YU

Dept. of Intelligent Robotics, SKKU

Outline

- Feature Extraction
- Introduction of PCA & LDA
- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (FLDA)
- Simple Enhancement of PCA/LDA

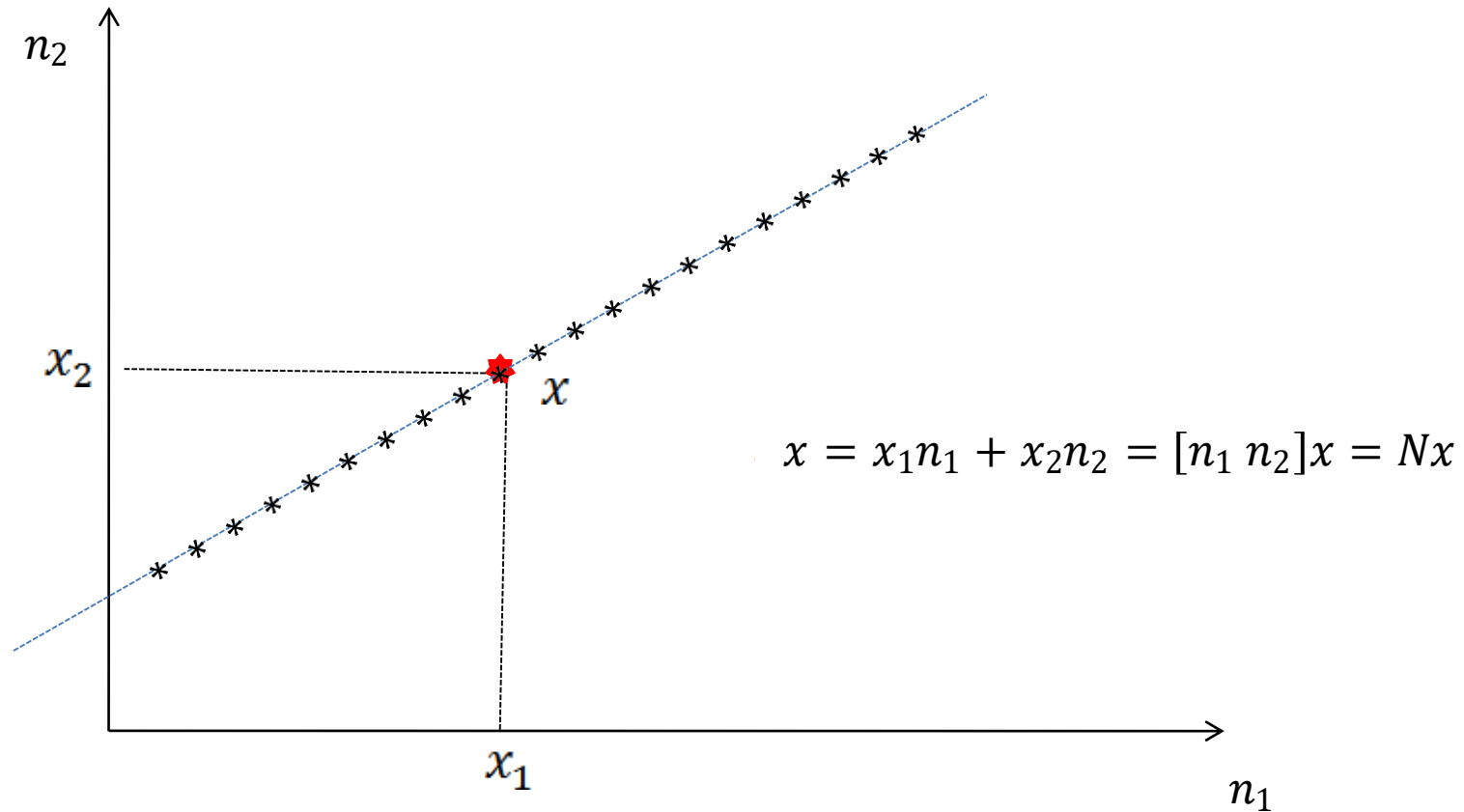
Feature Extraction

- Features
 - Weight, Height, Width, Volume, Head size, ...
 - Edge, Shape, Geometric Relations ...
 - RGB Color for each pixel
 - SIFT, SURF, HOG, ...
- Feature Extraction from Raw Data
 - Pixel Valued Vector is raw data vector
 - Raw data vector is redundant
 - The dimension should be reduced

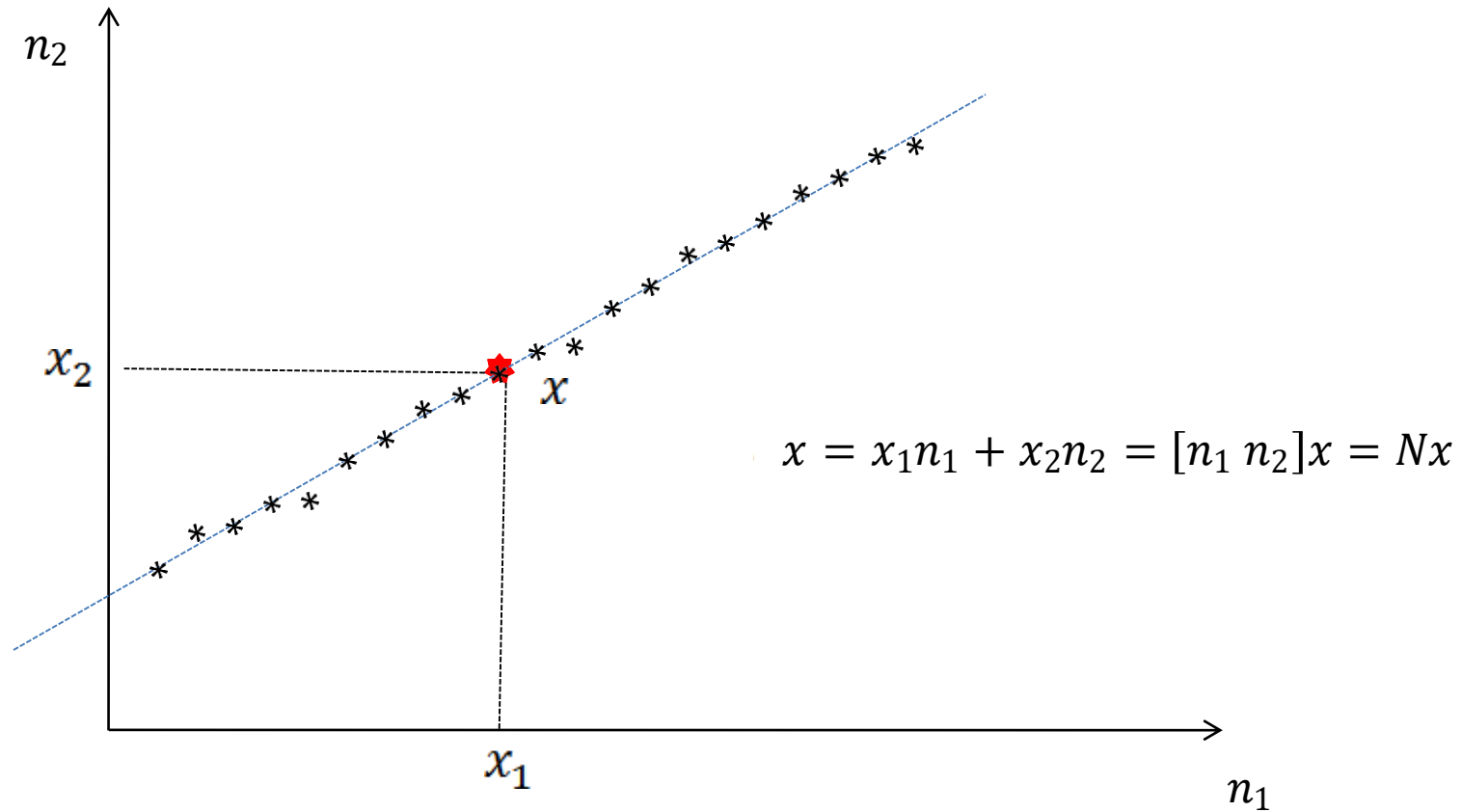
Component Analysis and Discriminants

- How to reduce excessive dimensionality?
 - Answer: Combine features highly dependent to each other.
- Linear methods project high-dimensional data onto lower dimensional space.
- Principal Components Analysis (PCA)
 - seeks the projection which best represents the data in a least-square error sense.
- Linear Discriminant Analysis (LDA) or Fisher Linear Discriminant
 - seeks the projection that best separates the data in a least-square discrimination error sense.

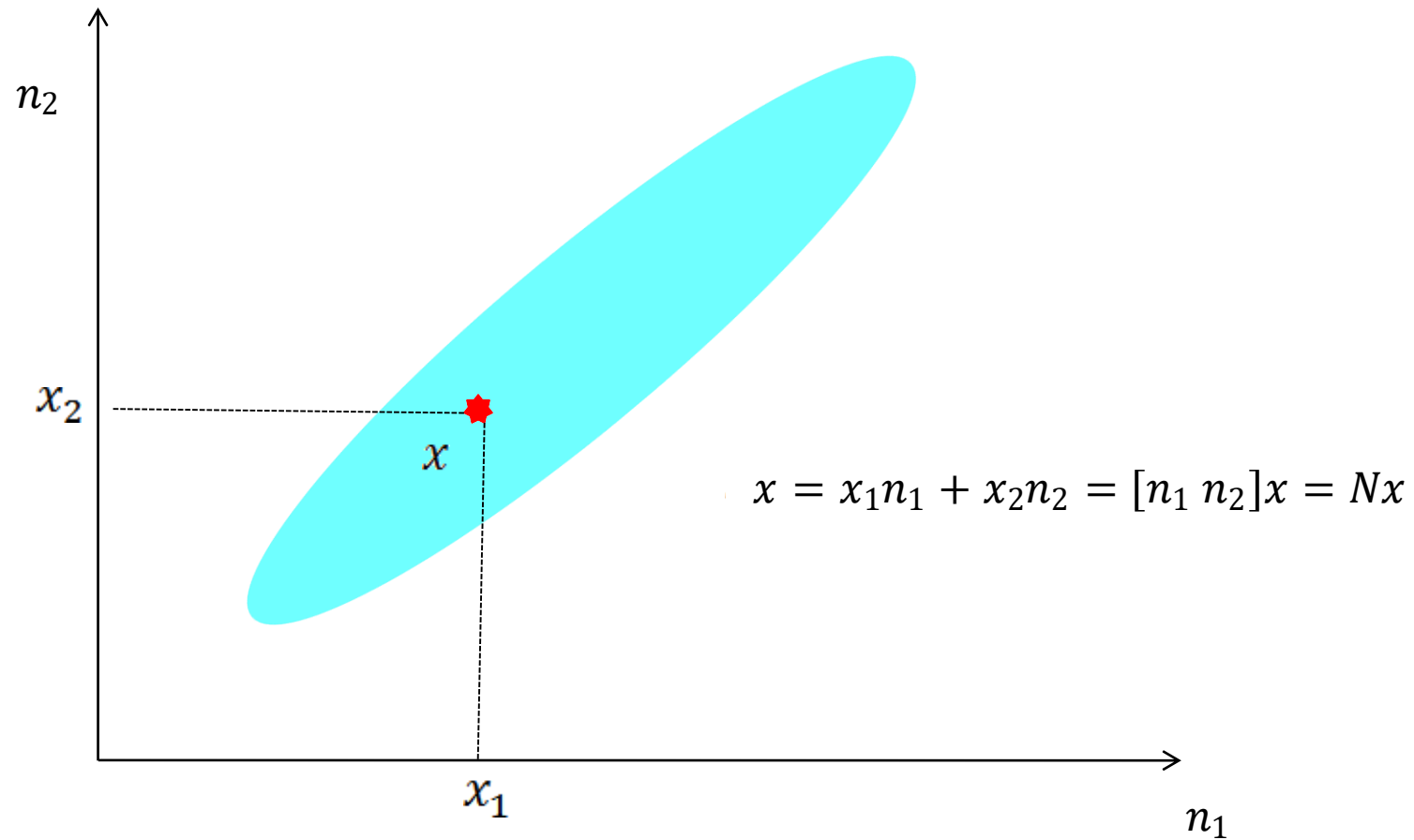
Principal Component Analysis



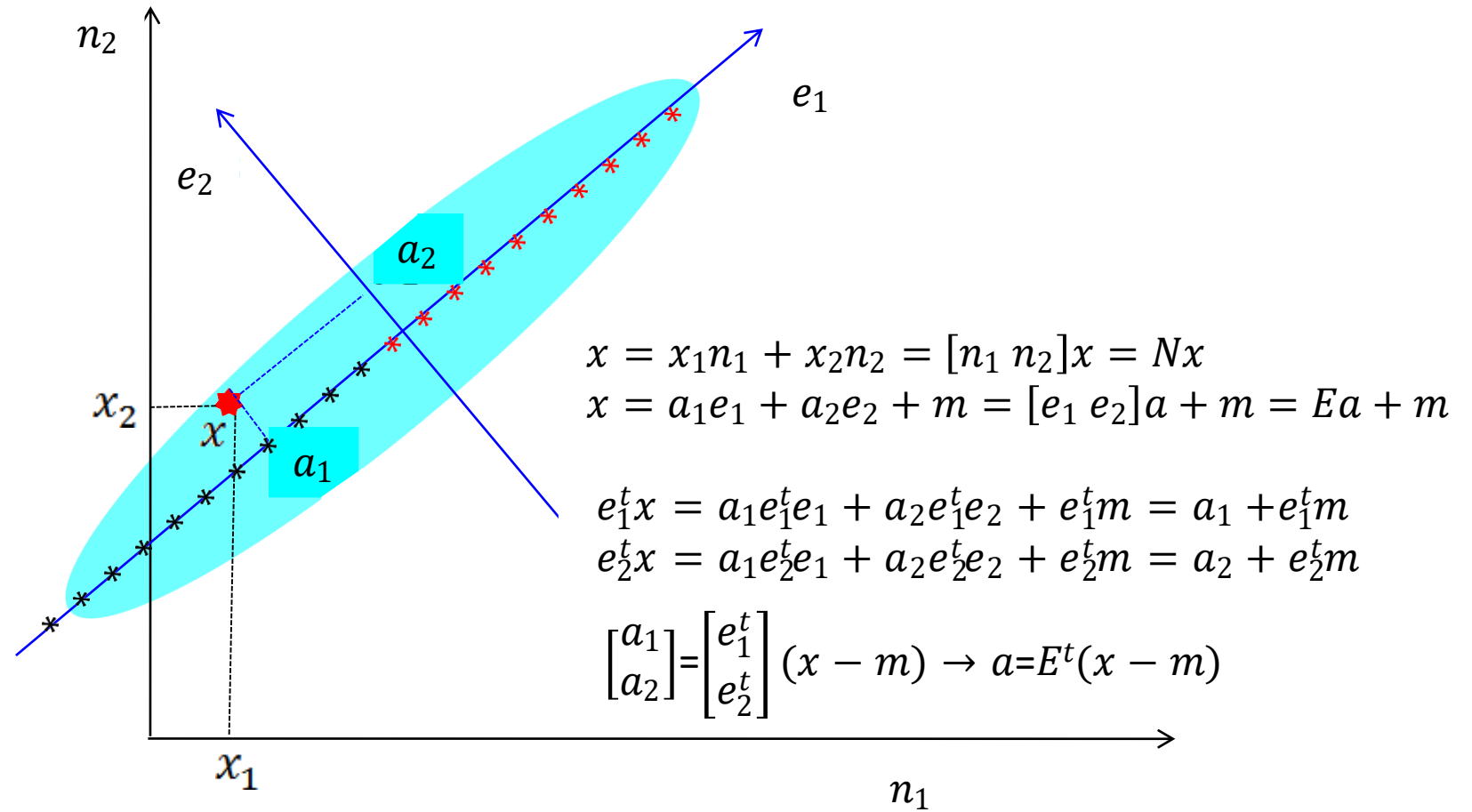
Principal Component Analysis



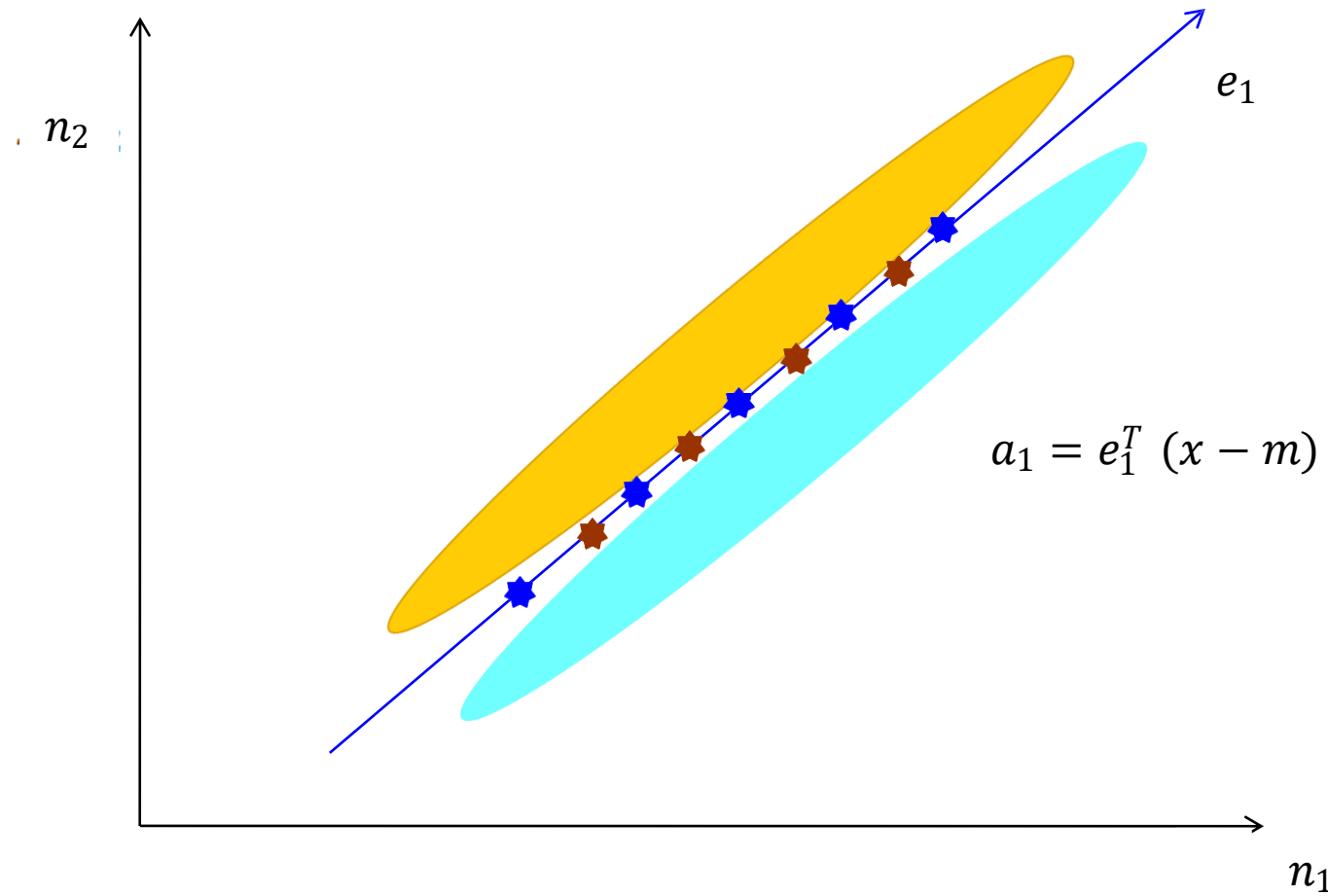
Principal Component Analysis



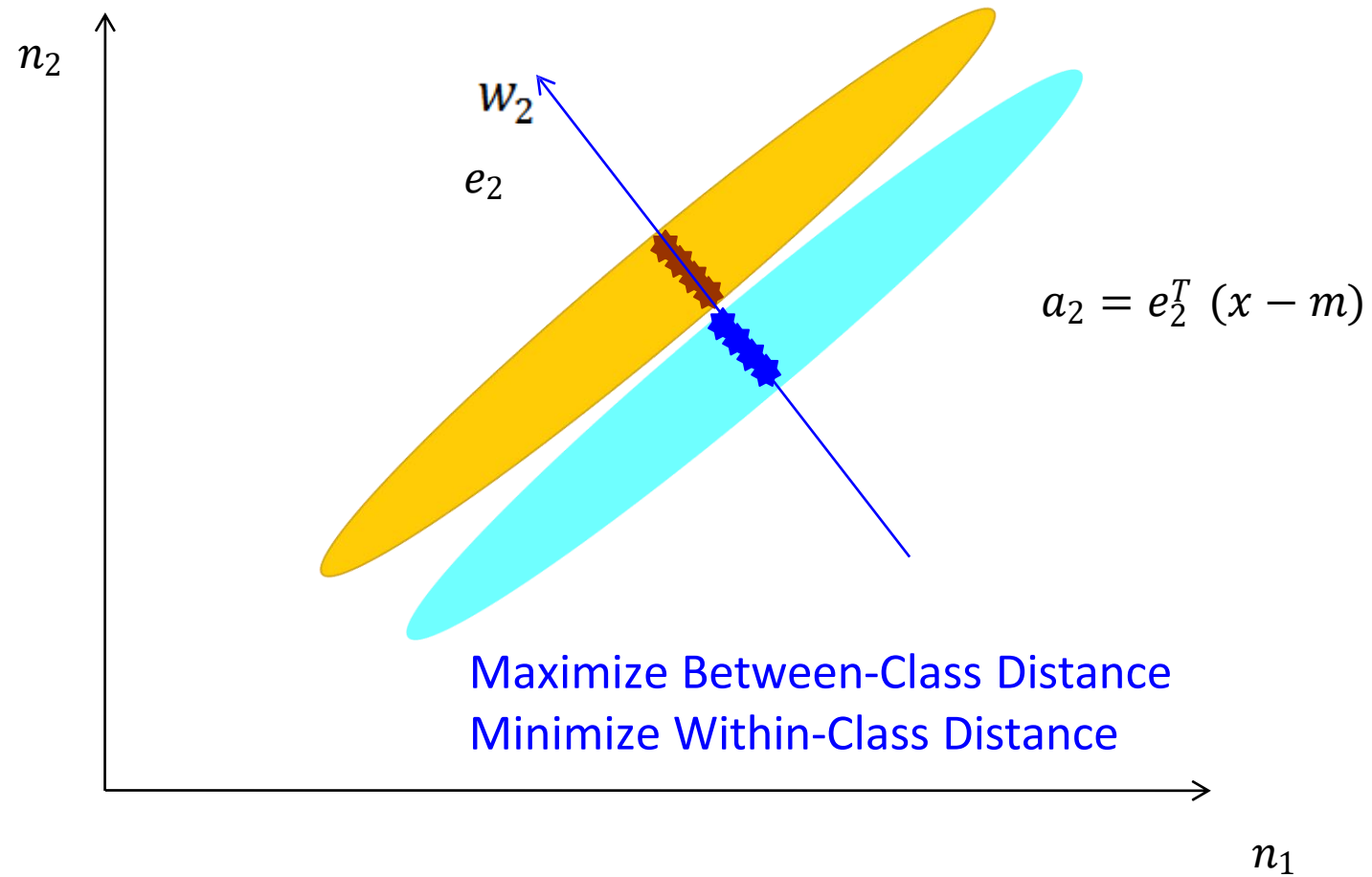
Principal Component Analysis



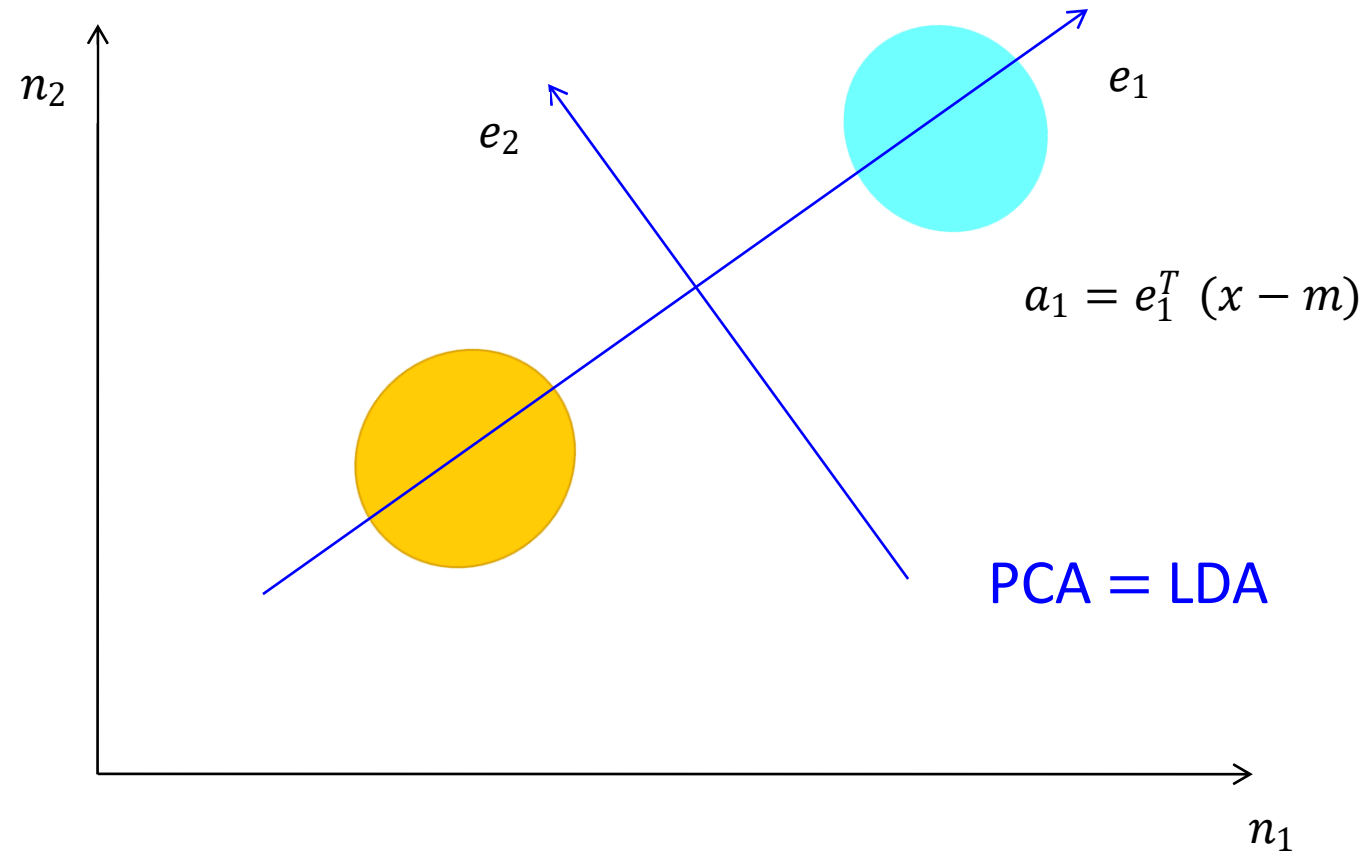
Linear Discriminant Analysis



Linear Discriminant Analysis



PCA & LDA



Principal Components Analysis (PCA)

- How to represent n d -dimensional vector samples $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ by a single vector $\mathbf{x}_0$?
 - Find $\mathbf{x}_0$ that minimizes squared error correction function

$$J_0(\mathbf{x}_0) = \sum_{k=1}^n \|\mathbf{x}_0 - \mathbf{x}_k\|^2.$$

Principal Components Analysis (PCA)

- How to represent n d -dimensional vector samples $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ by a single vector $\mathbf{x}_0$?

- Find $\mathbf{x}_0$ that minimizes squared error correction function

$$J_0(\mathbf{x}_0) = \sum_{k=1}^n \|\mathbf{x}_0 - \mathbf{x}_k\|^2.$$

- The solution is sample mean

$$\mathbf{x}_0 = \mathbf{m} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

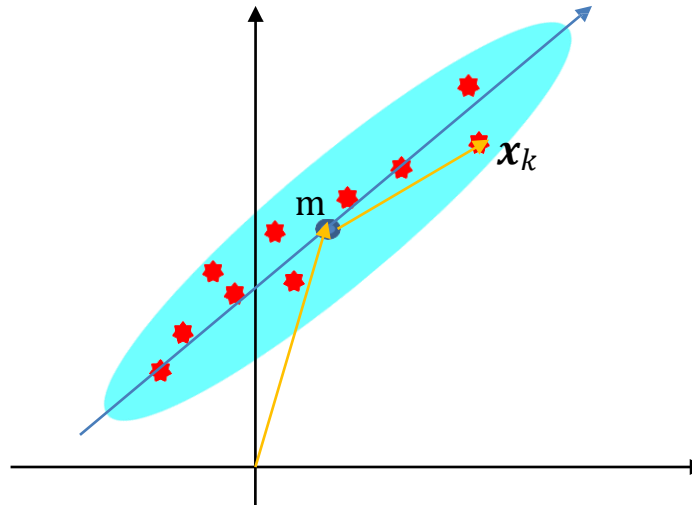
- This is **zero-dimensional** representation of the data set.
- **One-dimensional** representation by projecting the data onto a line through the sample mean reveals variability in the data.

Principal Components Analysis (PCA)

- This is **zero-dimensional** representation of the data set.

$$x_0 = m = \frac{1}{n} \sum_{k=1}^n x_k$$

- **One-dimensional** representation by projecting the data onto a line through the sample mean reveals variability in the data.



PCA ; Projection

- Let $\mathbf{e}$ be a unit vector in a direction of the line. The equation of the line

$$\mathbf{x} = \mathbf{m} + a\mathbf{e} \leftrightarrow \mathbf{e}^t(\mathbf{x} - \mathbf{m}) = a$$

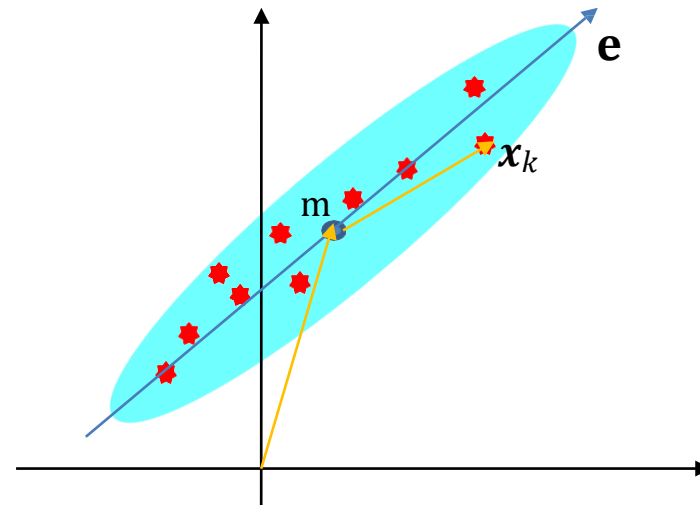
- Representing $\mathbf{x}_k$ by $\mathbf{m} + a_k\mathbf{e}$, find “optimal”

set minimizing criterion function :

$$J_1(a_1, \dots, a_n, \mathbf{e}) = \sum_{k=1}^n ||\mathbf{m} + a_k\mathbf{e} - \mathbf{x}_k||^2.$$

from $\partial J_1 / \partial a_k = 0$

we find $a_k = \mathbf{e}^t(\mathbf{x}_k - \mathbf{m})$



PCA ; Projection

- Representing $\mathbf{x}_k$ by $\mathbf{m} + a_k \mathbf{e}$, find “optimal” a_k

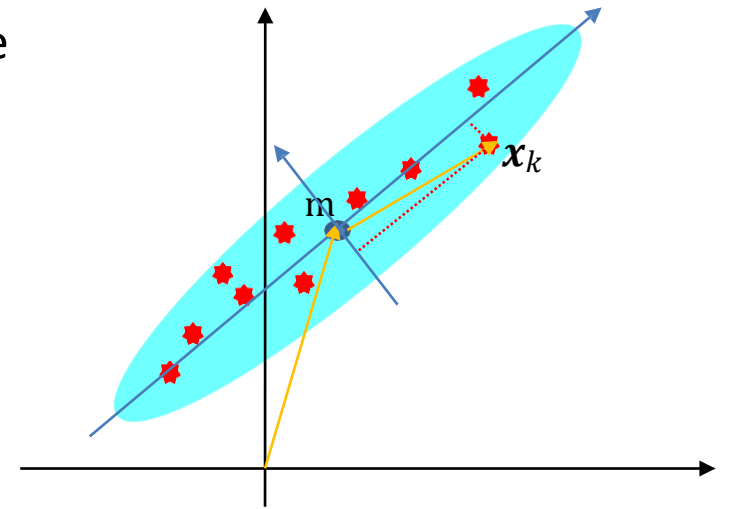
$$a_k = \mathbf{e}^T (\mathbf{x}_k - \mathbf{m})$$

- How to find the *best* direction for $\mathbf{e}$?
- The least square solution:** project the vector $\mathbf{x}_k$ onto the line in the direction of $\mathbf{e}$, passing through the sample mean.

$$J_1(a_1, \dots, a_n, \mathbf{e}) = \sum_{k=1}^n \|\mathbf{m} + a_k \mathbf{e} - \mathbf{x}_k\|^2.$$

$$a_k = \mathbf{e}^T (\mathbf{x}_k - \mathbf{m})$$

- Minimize J w.r.t $\mathbf{e}$.



PCA ; Scatter matrix

- Substituting a_k into $J_1(a, \mathbf{e})$ we find

$$\begin{aligned} J_1(a, \mathbf{e}) &= \sum_{k=1}^n a_k^2 \|\mathbf{e}\|^2 - 2 \sum_{k=1}^n a_k \mathbf{e}^T (\mathbf{x}_k - \mathbf{m}) + \sum_{k=1}^n \|\mathbf{x}_k - \mathbf{m}\|^2 \\ &= \sum_{k=1}^n a_k^2 - 2 \sum_{k=1}^n a_k^2 + \sum_{k=1}^n \|\mathbf{x}_k - \mathbf{m}\|^2 \\ &= -\sum_{k=1}^n [\mathbf{e}^T (\mathbf{x}_k - \mathbf{m})]^2 + \sum_{k=1}^n \|\mathbf{x}_k - \mathbf{m}\|^2 \\ &= -\sum_{k=1}^n \mathbf{e}^T (\mathbf{x}_k - \mathbf{m})(\mathbf{x}_k - \mathbf{m})^T \mathbf{e} + \sum_{k=1}^n \|\mathbf{x}_k - \mathbf{m}\|^2 \\ &= -\mathbf{e}^T \mathbf{S} \mathbf{e} + \sum_{k=1}^n \|\mathbf{x}_k - \mathbf{m}\|^2 \end{aligned}$$

where a *scatter matrix* $\mathbf{S}$ which is $(n - 1)$ times of sample covariance matrix given as

$$\mathbf{S} = \sum_{k=1}^n (\mathbf{x}_k - \mathbf{m})(\mathbf{x}_k - \mathbf{m})^T.$$

PCA ; Scatter matrix

- $J_1(a, \mathbf{e}) = -\mathbf{e}^T \mathbf{S} \mathbf{e} + \sum_{k=1}^n ||\mathbf{x}_k - \mathbf{m}||^2$
- Vector $\mathbf{e}$ that minimizes J_1 also maximizes $\mathbf{e}^T \mathbf{S} \mathbf{e}$.
- So we find $\mathbf{e}$, which maximize $\mathbf{e}^T \mathbf{S} \mathbf{e}$

subject to constraint $||\mathbf{e}||=1$

- Let λ be Lagrange multiplier.
- Differentiating L with respect to $\mathbf{e}$: $L = \mathbf{e}^T \mathbf{S} \mathbf{e} - \lambda(\mathbf{e}^T \mathbf{e} - 1)$

$$\nabla_{\mathbf{e}} L(\mathbf{e}) = 2\mathbf{S} \mathbf{e} - 2\lambda \mathbf{e}$$

- By setting to zero we see that $\mathbf{e}$ is an eigenvector of $\mathbf{S}$:

$$\mathbf{S} \mathbf{e} = \lambda \mathbf{e} \rightarrow \mathbf{e}^T \mathbf{S} \mathbf{e} = \lambda$$

- So to maximize $\mathbf{e}^T \mathbf{S} \mathbf{e}$ takes maximal λ_i . $\mathbf{e}_i$ should be normalized to each other.

PCA ; Scatter matrix

- The result is easily extended to d' dimensional projection:

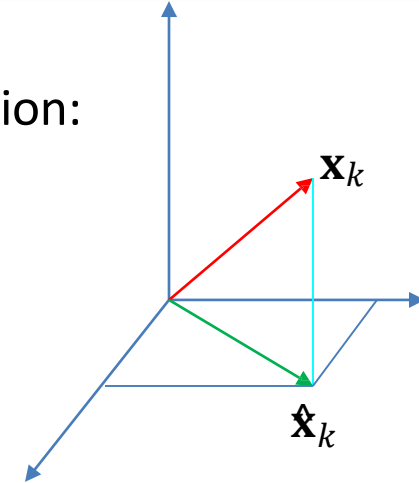
$$\hat{\mathbf{x}}_k = \mathbf{m} + \sum_{i=1}^{d'} a_k^i \mathbf{e}_i \quad \text{where} \quad d' \leq d$$

- The criterion function

$$J_{d'} = \sum_{k=1}^n \left\| \left(\mathbf{m} + \sum_{i=1}^{d'} a_k^i \mathbf{e}_i \right) - \mathbf{x}_k \right\|^2$$

is minimized when $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{d'}$ are the eigenvectors having the largest eigenvalues.

- The coefficients $a_k^i = \mathbf{e}_i^T (\mathbf{x}_k - \mathbf{m})$ are *principal components*.



Error function

- If $d' < d$ error which is made by dropping the last terms is

$$\begin{aligned} J_{d'} &= \sum_{k=1}^n \left\| \sum_{i=d'+1}^d a_k^i \mathbf{e}_i \right\|^2 \\ &= \sum_{i=d'+1}^d \mathbf{e}_i^T \sum_{k=1}^n (\mathbf{x}_k - \mathbf{m})(\mathbf{x}_k - \mathbf{m})^T \mathbf{e}_i \\ &= \sum_{i=d'+1}^d \mathbf{e}_i^T \mathbf{S} \mathbf{e}_i = \sum_{i=d'+1}^d \lambda_i \end{aligned}$$

$$\mathbf{x}_k = \mathbf{m} + \sum_{i=1}^{d'} a_k^i \mathbf{e}_i$$

$$a_k^i = \mathbf{e}_i^T (\mathbf{x}_k - \mathbf{m})$$

- This is a sum of lowest eigenvalues.

PCA – the algorithm

- Input: $X^{(n)} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, $\mathbf{x}_k = (x_1^k, \dots, x_d^k)$
- Take $d' < d$
- Output: $A^{(n)} = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ $\mathbf{a}_k = (a_k^1, \dots, a_k^{d'})$
- Algorithm:
 - Compute the mean of the training set $\mathbf{m} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$.
 - Compute the scatter matrix $\mathbf{S}$.
 - Find eigenvectors of $\mathbf{S}$ and corresponding eigenvalues:
$$S\{\mathbf{e}_i, \lambda_i\}_{i=1}^d, \quad \forall i: \mathbf{S}\mathbf{e}_i = \lambda_i \mathbf{e}_i, \lambda_1 \geq \lambda_2 \geq \dots \lambda_d$$
 - Choose d' eigenvectors, and for each sample $\mathbf{x}_k$ point compute
$$a_k^i = \mathbf{e}_i^T (\mathbf{x}_k - \mathbf{m}), \quad i = 1, \dots, d'$$

Interim Summary

- Principal Component Analysis
 - ✓ Feature Extraction
 - ✓ Dimension Reduction

$$J_{d'} = \sum_{k=1}^n \left\| \left(\mathbf{m} + \sum_{i=1}^{d'} a_k^i \mathbf{e}_i \right) - \mathbf{x}_k \right\|^2$$

$$\mathbf{S} = \sum_{k=1}^n (\mathbf{x}_k - \mathbf{m})(\mathbf{x}_k - \mathbf{m})^T.$$

$$\mathbf{S}\mathbf{e} = \lambda \mathbf{e}$$

$$\mathbf{e}^T \mathbf{S} \mathbf{e} = \lambda$$

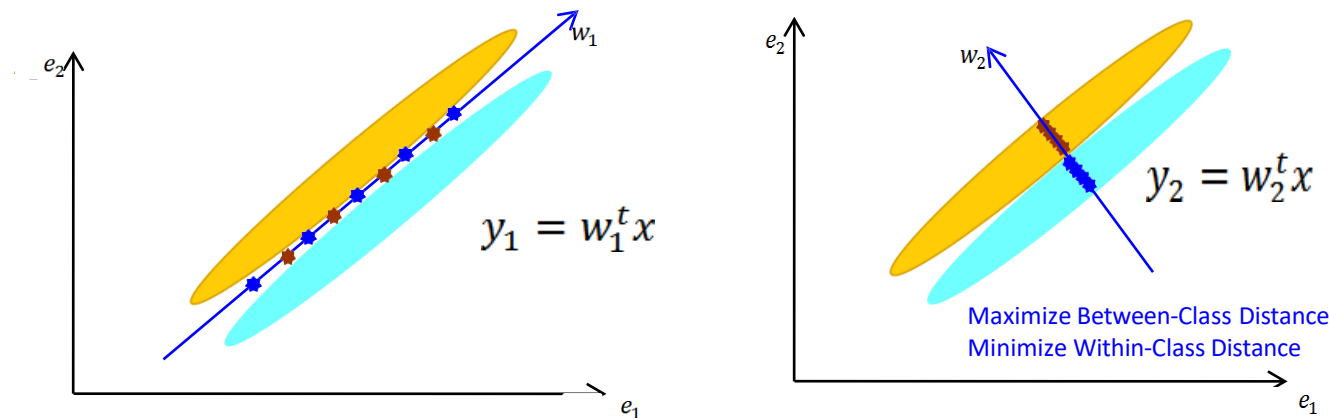
$$a_k^i = \mathbf{e}_i^T (\mathbf{x}_k - \mathbf{m}), i = 1, \dots, d'$$

$$\begin{bmatrix} a_k^1 \\ a_k^2 \\ \vdots \\ a_k^{d'} \end{bmatrix} = \begin{bmatrix} e_1^T \\ e_2^T \\ \vdots \\ e_{d'}^T \end{bmatrix} (\mathbf{x}_k - \mathbf{m})$$

$$\mathbf{a}_k = E^T (\mathbf{x}_k - \mathbf{m})$$

Linear Discriminant Analysis: LDA

- We have n d -dimensional samples $x_1, \dots, x_n$ in a subset D_1 , labeled w_1 and n_2 in a subset D_2 , labeled w_2 .
- Find direction of line w , that maximally separate the data.



- Let a difference between sample means be a measure of separation of projected points

Fisher Linear Discriminant

- Project samples $\mathbf{x}_k$ onto $\mathbf{w}$.

$$y_k = \mathbf{w}^t \mathbf{x}_k$$

- n samples y_k are divided into the subsets Y_1 and Y_2
- Let $\mathbf{m}_i$ be the sample mean

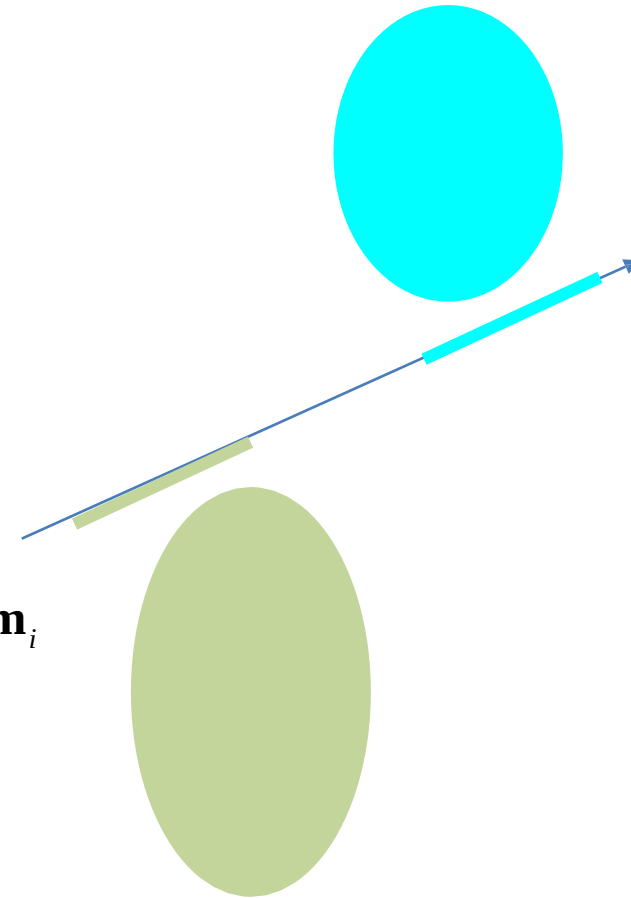
$$\mathbf{m}_i = \frac{1}{n_i} \sum_{\mathbf{x} \in D_i} \mathbf{x}$$

- The sample mean for projected points

$$\tilde{m}_i = \frac{1}{n_i} \sum_{y \in Y_i} y = \frac{1}{n_i} \sum_{\mathbf{x} \in D_i} \mathbf{w}^t \mathbf{x} = \mathbf{w}^t \mathbf{m}_i$$

- Distance between the projected means is

$$| \tilde{m}_1 - \tilde{m}_2 | = | \mathbf{w}^t (\mathbf{m}_1 - \mathbf{m}_2) |$$



Fisher Linear Discriminant

- A scatter for projected samples labeled ω_i

$$\tilde{s}_i^2 = \sum_{y \in Y_i} (y - \tilde{m}_i)^2$$

$(1/n)(\tilde{s}_1^2 + \tilde{s}_2^2)$ is an **estimate of the variance** of the pooled data.

$\tilde{s}_1^2 + \tilde{s}_2^2$ is called **total within-class scatter** of the projected samples.

- The Fisher discriminant employs $\mathbf{w}^t \mathbf{x}$ for which criterion

$$J(\mathbf{w}) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

is maximum

Fisher Linear Discriminant

- Define scatter matrices S_i and S_w by

$$S_i = \sum_{\mathbf{x} \in D_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^t$$

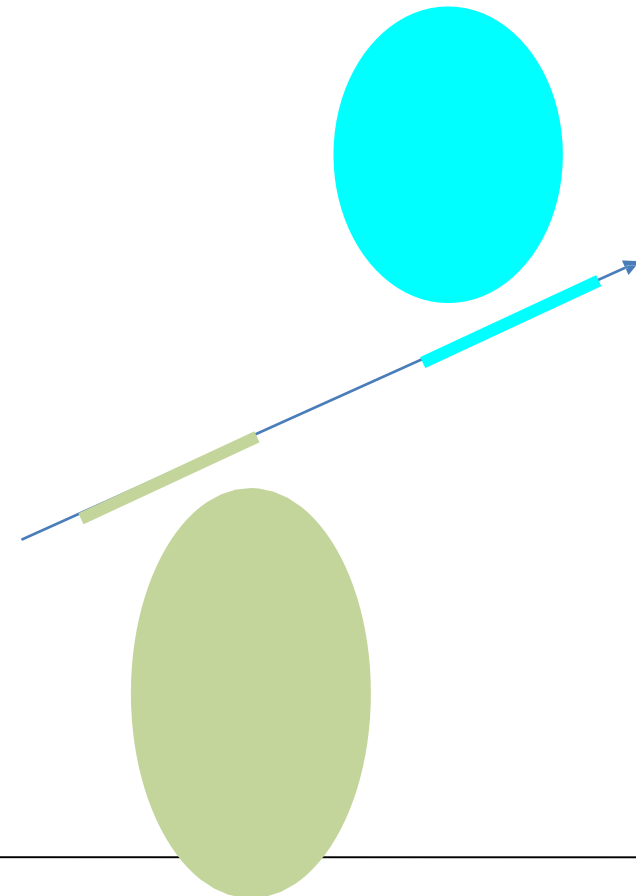
and

$$S_w = S_1 + S_2$$

- Then

$$\tilde{s}_i^2 = \sum_{\mathbf{x} \in D_i} (\mathbf{w}^t \mathbf{x} - \mathbf{w}^t \mathbf{m}_i)^2 = \sum_{\mathbf{x} \in D_i} \mathbf{w}^t (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^t \mathbf{w} = \mathbf{w}^t S_i \mathbf{w}$$

- Thus $\tilde{s}_1^2 + \tilde{s}_2^2 = \mathbf{w}^t S_w \mathbf{w}$



Fisher Linear Discriminant

- Similarly,

$$(\tilde{m}_1 - \tilde{m}_2)^2 = (\mathbf{w}^t \mathbf{m}_1 - \mathbf{w}^t \mathbf{m}_2)^2 = \mathbf{w}^t (\mathbf{m}_1 - \mathbf{m}_2) (\mathbf{m}_1 - \mathbf{m}_2)^t \mathbf{w} = \mathbf{w}^t \mathbf{S}_B \mathbf{w}$$

$\mathbf{S}_w$ is called **within-class scatter matrix** (proportional to sample covariance matrix)

$\mathbf{S}_B = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^t$ is called **between-class scatter matrix**.

- This gives the equivalent expression for Fisher's discriminant

$$J(\mathbf{w}) = \frac{\mathbf{w}^t \mathbf{S}_B \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}} = \lambda$$

- Which vector $\mathbf{w}$ maximizes it?

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{2\mathbf{S}_B \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}} - \frac{\mathbf{w}^t \mathbf{S}_B \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}} \frac{2\mathbf{S}_W \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}} = 0 \rightarrow \mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w}$$

Fisher Linear Discriminant

- Hence one gets

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w}, \quad \lambda = \frac{\mathbf{w}^t \mathbf{S}_B \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}},$$

or equivalently

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w},$$

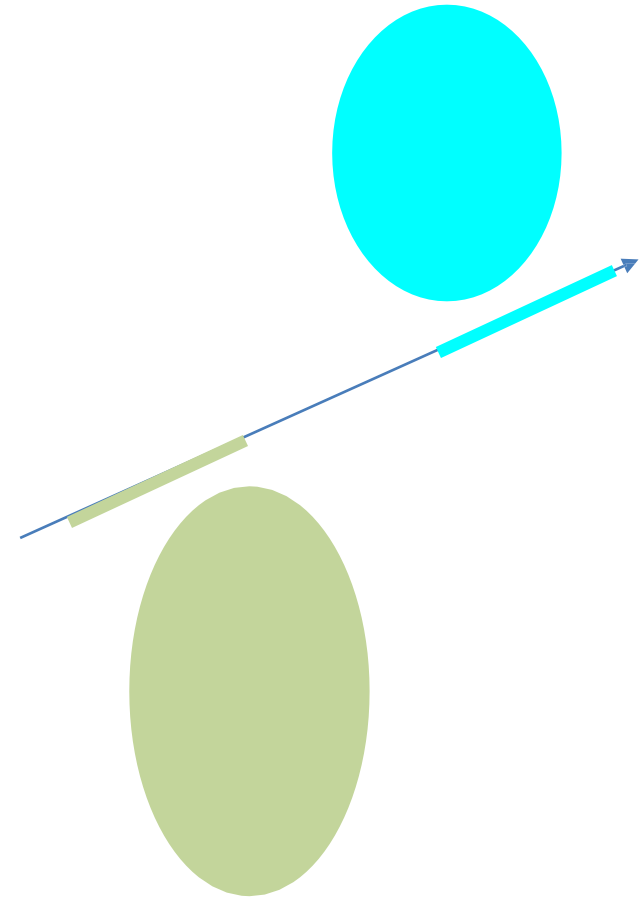
- Since for any $\mathbf{w}$, $\mathbf{S}_B \mathbf{w}$ is always in the direction of $\mathbf{m}_1 - \mathbf{m}_2$:

$$\mathbf{S}_B \mathbf{w} = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^t \mathbf{w} = \alpha(\mathbf{m}_1 - \mathbf{m}_2)$$

- It is not necessary to determine the eigenvalues of $\mathbf{S}_W^{-1} \mathbf{S}_B$.
- One simply gets

$$\mathbf{w} \propto \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

- Scale factor for $\mathbf{w}$ is unimportant (why?).
- FLDA is one-dimensional projection



Summary

- Feature Extraction
- Introduction of PCA & LDA
- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (FLDA)
- Simple Enhancement of PCA/LDA